InterviewLens

AI-Powered Real Interview Analysis Platform

Architecture designed for the micro1 Agentic Workflows Hackathon, focused on authorized interview-data ingestion, agentic analysis, evidence verification, human review, and longitudinal interview insights.

1. Overall Architecture

The meeting platform is treated as a data source. The core product value is the agentic analysis, verification, expert feedback, and cross-interview learning loop.

```
USER
|
v
Authentication (Google / Microsoft / GitHub)
|
v
User Profile (Resume, Skills, Interview History)
|
v
Create Interview (Job Description + Meeting URL)
|
v
DATA INGESTION
|---- Google Meet (authorized API)
|---- Microsoft Teams (authorized API)
|---- Zoom (authorized API)
|---- Transcript upload
|---- Authorized recording upload
|
v
Transcript Normalizer
|
v
Interview Orchestrator
|---- Technical Agent
|---- Communication Agent
|---- Behavioral Agent
|
v
Evidence / Verification Agent
|
v
Human / Expert Reviewer
|
v
Verified Interview Assessment
|
v
Interview Memory
|
v
Cross-Interview Analysis
|
v
Personalized Insights & Next Preparation
```

2. Authentication & Authorization

Use OAuth/OIDC-based authentication rather than asking users to type an email address as proof of identity. Authentication establishes who the user is; authorization determines what interview data the application is allowed to access.

Flow: Google/Microsoft/GitHub login → verified application account → explicit meeting/data permissions → authorized ingestion.

A successful login must not be treated as automatic permission to access every meeting belonging to that account.

3. Interview Creation

The user creates an interview record containing the interview name, resume, job description, meeting platform, meeting URL/identifier, and ingestion method. Suggested lifecycle: WAITING_FOR_INTERVIEW → INTERVIEW_COMPLETED → TRANSCRIPT_AVAILABLE → ANALYZING → REVIEW_REQUIRED → COMPLETED.

4. Data Ingestion Layer

The system should not assume that every meeting URL can be captured automatically. It should use official, authorized platform access when available and provide an upload fallback when it is not.

Source	Preferred path	Fallback
Google Meet	Authorized API / transcript	User-uploaded transcript or authorized recording
Microsoft Teams	Authorized Graph access	User-uploaded transcript or authorized recording
Zoom	Authorized API / transcript	User-uploaded transcript or authorized recording

5. Transcript Normalizer

Different platforms return different transcript structures. Normalize them into one internal representation containing interview ID, speaker, timestamp, and text. Downstream agents should not need to know whether the source was Meet, Teams, Zoom, or an uploaded file.


```
{"interview_id":"123", "segments":[{"speaker":"interviewer","timestamp":"00:02:15","text":"Explain your
project."},{"speaker":"candidate","timestamp":"00:02:24","text":"I built a..."}]}
```


6. Agentic Analysis Layer

The Interview Orchestrator decides which specialized analyses are required. Specialized agents are used because different dimensions of interview performance require different evaluation logic.

Technical Agent

Evaluates question understanding, correctness, expected concepts, concepts demonstrated, missing concepts, and supporting evidence.

Communication Agent

Evaluates observable signals such as answer relevance, structure, conciseness, repetition, filler words, long pauses, clarification requests, interviewer follow-ups, and incomplete answers.

Behavioral Agent

Evaluates behavioral answers using structure such as Situation, Task, Action, Result, specificity, ownership, and outcomes.

7. Evidence / Verification Agent

Every important AI claim should be checked against the original transcript. The verifier locates supporting questions/answers and can accept, modify, or reject an assessment. This reduces unsupported statements such as 'you lack confidence' when the available evidence only shows observable communication behavior.

```
Claim → find transcript evidence → verify → accept / modify / reject
```

8. Human / Expert Review

An experienced interviewer can review the AI assessment, agree, modify, or reject findings, and provide comments. The final report combines AI analysis with qualified human feedback. The system should be positioned as an assessment/coaching tool rather than an automated hiring decision-maker.

9. Interview Memory & Cross-Interview Analysis

Store structured insights from multiple interviews, not merely raw transcripts. The Cross-Interview Agent identifies recurring weaknesses, improving areas, repeated technical gaps, communication patterns, and role-specific issues.

```
Interview 1 + Interview 2 + Interview 3 + ... → Interview Memory → Cross-Interview Agent → Recurring
Patterns
```

10. Knowledge Gap vs. Interview Performance Gap

A key differentiator is distinguishing lack of knowledge from difficulty retrieving or communicating knowledge under interview conditions. After an interview, the system can re-test questions the candidate struggled with and compare the results.

```
During interview: Redis 5/10 → Post-interview test: 9/10 → likely performance/retrieval issue.
During interview: System Design 4/10 → Post-interview test: 4/10 → likely knowledge gap.
```

11. Final Dashboard

Dimension	Example result
Technical Knowledge	8.2 / 10
Communication	6.1 / 10
Answer Structure	5.4 / 10
Problem Solving	7.8 / 10
Behavioral Answers	6.3 / 10
Question Understanding	7.5 / 10

The dashboard should show recurring weaknesses with evidence, improvements over time, possible performance gaps, and prioritized preparation actions.

12. Practical Backend Architecture

A practical implementation can use a React/Next.js frontend, a Node.js API for application/authentication/data management, and a Python AI service for the agent workflow. PostgreSQL can hold structured application data, Redis can support queues/cache/memory, and object storage can hold authorized uploaded artifacts.

```
React / Next.js
|
v
API Gateway / Node.js API
|
+---- Authentication & OAuth
+---- Interview management
+---- File metadata
+---- Expert review
|
v
Python AI Service
|
+---- Orchestrator
+---- Technical Agent
+---- Communication Agent
+---- Behavioral Agent
+---- Verification Agent
+---- Cross-Interview Agent
|
+---- PostgreSQL
+---- Redis
+---- Object Storage
```

13. End-to-End Agent Workflow

```
REAL INTERVIEW
↓
Authorized Capture / Upload
↓
Transcript Normalization
↓
Interview Orchestrator
↓
Technical + Communication + Behavioral Agents
↓
Evidence Verification
↓
Human Expert Review
↓
Verified Assessment
↓
Interview Memory
↓
Cross-Interview Analysis
↓
Personalized Insights
↓
Next Preparation
↓
Next Interview
■
```

14. How This Matches the micro1 Hackathon

Hackathon criterion	InterviewLens mapping
Problem & User Value	Students/job seekers struggle to understand why preparation does not translate into real interview performanc
Agent Solution & Engineering	Orchestration, specialized analysis agents, verification, memory, and human review.
End-to-End Quality	Authorized interview data → analysis → verified report → actionable preparation.
Measured Improvement	Compare a simple LLM baseline against the multi-agent workflow on the same evaluation cases.
Reproducibility	Fixed evaluation dataset, setup instructions, commands, versions, and expected outputs.

Hot Take / Insight	A poor interview answer is not always a knowledge gap; it may be a retrieval, structure, or communication pro

15. Baseline vs. Final Agent Solution

Baseline: Transcript → one general LLM prompt → generic interview feedback.

Final: Transcript → orchestrator → specialized agents → evidence verification → expert review → interview memory → cross-interview insights.

The same evaluation cases should be used for both approaches. The hackathon recommends defining a primary metric before evaluation and targeting ten or more cases when the task allows it.

16. Recommended Hackathon MVP

• Google login / OAuth

• Resume upload

• Job-description upload

• Interview transcript upload

• Question/answer extraction

• Technical analysis agent

• Communication analysis agent

• Evidence verification agent

• Human expert review

• Final interview report

• Multiple-interview storage

• Cross-interview recurring weakness detection

• Knowledge-vs-performance re-test

17. Integration Roadmap

Version	Focus
V1	Transcript upload + agentic analysis + verification + expert review
V2	Multiple interviews + longitudinal memory + recurring pattern detection
V3	Google Meet / Teams / Zoom authorized ingestion adapters
V4	Real-time transcription where platform permissions and consent support it

18. Core Product Principle

Do not make Google Meet, Teams, or Zoom the heart of the product. Treat them as input adapters. The heart of InterviewLens is the loop:

> REAL INTERVIEW DATA → AGENTIC ANALYSIS → VERIFICATION → HUMAN FEEDBACK → INTERVIEW MEMORY → CROSS-INTERVIEW INSIGHT → BETTER PREPARATION

Privacy principle: only process interview data that the user is authorized to provide. Do not bypass meeting-platform or organizational controls. Keep credentials and private information outside the submission, as required by the hackathon rules.